

Itbaan Safwan

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EDUCATION

B.S. Computer Science

Institute of Business Administration

Karachi, Pakistan

June 2026

- **CGPA: 3.47 / 4.00.** Relevant coursework: Neural Networks and Deep Learning, Introduction to Natural Language Processing, Computer Vision, Linear Algebra, Numerical Analysis.
- Final year project: *Visual Question Answering for Gastrointestinal Imaging with Multimodal Reasoning*, advised by Dr. Muhammad Atif Tahir.

RESEARCH EXPERIENCE

Independent Research Project

2026

Model internals study | *Technical write-up*

- Replicated prior visual-token analysis on **170 COCO images** and built a controlled activation-patching study on **150 synthetic samples** spanning five perceptual tasks in the 3B-parameter LFM2.5-VL; 146 clean/counterfactual pairs were answered correctly and used for causal analysis.
- Found **task-specific causal readout depths** across the model's 30 layers: spatial relation peaked at L5 (0.074), counting at L13 (0.268), and orientation, shape, and colour at L21 (0.324, 0.243, and 0.313 recovery drop respectively).
- Patching the changed object produced best recoveries of **0.185–0.733** versus only **0.001–0.013** for other objects in the same scene, a **47×–933× specificity gap**; recovery shifted only when crossing the model's eight attention layers and vanished after the final attention layer.

Institute of Business Administration

Lead researcher | *Supervisor: Dr. Muhammad Atif Tahir*

Karachi, Pakistan

2025 – 2026

- **First-authored** *Towards Grounded GI Endoscopy VQA via Multi-Task Learning on Small VLMs*, accepted for a **poster presentation at the EMA4MICCAI Workshop, MICCAI 2026**; studied whether auxiliary visual supervision helps small VLMs answer from the correct visual evidence.
- Built auxiliary supervision without new manual annotation by deriving Grad-CAM localisation from a domain-pretrained GastroNet-5M classifier and generating visual descriptions with Gemma-27B restricted to observable image features.
- Fine-tuned **three VLM backbones** (Florence-2, Qwen3.5, InternVL3.5) jointly on localisation, description and VQA with QLoRA on a single RTX 3090 Ti; auxiliary tasks were used only during training, adding **no inference-time cost**.
- Improved VQA accuracy across all three backbones; Florence-2 increased from **79.26% to 82.00%**, with the largest gains on appearance, spatial, and higher-complexity questions.
- Evaluated grounding directly from internal token-to-patch representations; Florence-2 pointing-game accuracy improved from **0.333 to 0.805** in-distribution and **0.142 to 0.645** out-of-distribution (Wilcoxon signed-rank, $p = 1.47 \times 10^{-36}$).

MediaEval 2025 Medico Shared Task

Team lead | *Supervisor: Dr. Muhammad Atif Tahir*

Karachi, Pakistan (presented online)

2025

- Led the team submission *Multi-Task Learning for Visually Grounded Reasoning in Gastrointestinal VQA*, earning the **Distinctive Award and 2nd place in both subtasks** at MediaEval 2025 Medico.
- Fine-tuned a single Florence-2 backbone with LoRA across VQA, explanation generation, and localisation, using task-specific prompts so one model served all three objectives.
- Built two training sets: **3,344 generated explanations** with Gemma-27B and **4,337 text-mask pairs** from ClipSeg pseudo-masks plus expert masks from Kvasir-SEG and Kvasir-Instrument.
- Showed that joint training removed an adapter-capacity trade-off between language quality and localisation, increasing IoU from **0.2961 to 0.7403** at the same LoRA rank while BLEU also improved.
- Achieved **BLEU 0.4539, ROUGE-L 0.6531, CHRF++ 63.79, and BERTScore F1 0.9479** on the official private test set.

MAHED Shared Task, ArabicNLP 2025

Karachi, Pakistan

Sole author, independent project

2025

- Published a **sole-authored paper in the ACL Anthology** at ArabicNLP 2025 (co-located with EMNLP 2025), comparing multimodal fusion strategies for Arabic hateful-meme detection.
- Held CLIP ViT-B/32 and MARBERT backbones fixed while comparing cross-attention fusion, progressive 1-D convolutional fusion, and contrastive pre-training followed by gated classification.
- Implemented and trained all systems in PyTorch and addressed severe class imbalance with **weighted focal loss**.
- Conducted a controlled comparison of three multimodal fusion architectures, with progressive convolutional fusion achieving the strongest performance at **0.779 macro F1**.

PUBLICATIONS

1. **Safwan, I.**, Khan, R., Shaikh, M. A., & Tahir, M. A. (2026). Towards Grounded GI Endoscopy VQA via Multi-Task Learning on Small VLMs. *EMA4MICCAI Workshop, MICCAI 2026*. (accepted, poster presentation) [\[preprint\]](#)
2. **Safwan, I.**, Shaikh, M. A., Haaris, M., Khan, R., & Tahir, M. A. (2025). Multi-Task Learning for Visually Grounded Reasoning in Gastrointestinal VQA. *Proceedings of the MediaEval 2025 Multimedia Evaluation Workshop*, CEUR-WS. [\[paper\]](#)
3. **Safwan, I.** (2025). Thinking Nodes at MAHED: A Comparative Study of Multimodal Architectures for Arabic Hateful Meme Detection. *Proceedings of ArabicNLP 2025*, 712–719. Association for Computational Linguistics. [\[paper\]](#)

CONFERENCE PRESENTATIONS

- EMA4MICCAI Workshop, MICCAI 2026. **Poster presentation**, Oct 2026. *Towards Grounded GI Endoscopy VQA via Multi-Task Learning on Small VLMs*.
- MediaEval 2025 Multimedia Evaluation Workshop, Dublin and online. **Oral presentation**, Oct 2025. *Multi-Task Learning for Visually Grounded Reasoning in Gastrointestinal VQA*.

TEACHING EXPERIENCE

Institute of Business Administration

Karachi, Pakistan

Teaching Assistant, Theory of Automata

Jan 2026 – May 2026

- Led **weekly problem-solving tutorials** on automata, formal languages, and related course concepts, translating theory into worked examples and structured practice.
- Supported labs and office hours and collaborated with faculty on **assignment design, evaluation, and grading**.

Institute of Business Administration

Karachi, Pakistan

Teaching Assistant, Neural Networks and Deep Learning

Aug 2025 – Dec 2025

- Supported labs and office hours across neural-network theory and implementation, helping students connect mathematical concepts with practical deep-learning code.
- Collaborated with faculty on **assignment design, evaluation, and grading** and introduced current tooling into course material.

AI ENGINEERING EXPERIENCE

Cervana AI

Karachi, Pakistan

AI Research Intern

June 2026 – August 2026

- Benchmarked **9 Arabic TTS models** across WER, UTMOS naturalness, F0 prosody variation, and real-time factor; the selected model achieved **0.046 WER at 0.23x real time** using Whisper-Large-v3 for ASR evaluation.
- Fine-tuned an open-source TTS model for Urdu on AWS and redesigned it for streaming, reducing end-to-end latency from about **1000 ms to 200 ms** (approximately **5x faster**).
- Built and deployed the production-facing FastAPI service with voice cloning, API-key authentication, SQLite quotas, Docker, and an AWS EC2 GPU instance.

Systems Limited

AI Intern

Karachi, Pakistan
July 2025 – September 2025

- Built a stateful **agentic lead-generation system** with LangGraph and Gemini, using a ReAct loop to update lead state, maintain memory, and query internal data sources.
- Implemented a **multi-tier retrieval pipeline** across keyword, structured, and unstructured sources, ordered by priority to reduce unnecessary model calls.
- Prototyped a **CPU-only voice-to-voice agent** in Streamlit using Whisper, KokoroTTS, and real-time sockets.

LEADERSHIP AND OUTREACH

Google Developer Groups

Tech Lead for Generative AI

Karachi, Pakistan
2026

- Designed and delivered a two-stage generative AI workshop from first principles through hands-on experimentation, reaching **50+ students**.
- Structured the sessions to move from conceptual foundations to practical model use and experimentation.

IBA Computer Science and Data Science Society

Module Head, Machine Learning and NLP

Karachi, Pakistan
2025, 2026

- Led Machine Learning and NLP competition modules for **100+ participants** across 2025 and 2026, designing original problem statements that rewarded rigorous methodology rather than leaderboard tuning alone.
- Curated high-difficulty, **leakage-free datasets**, including a geopolitical NLP challenge built from structured news data.
- Evaluated submissions using both predictive performance and written methodology reports.